

# **AI-Based Stock Market Analysis and Recommendation System: A Data-Driven Approach to Investment**

## **Decision Support**

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### **Abstract**

The scope of this capstone project is to build an AI-Based Stock Market Analysis and Recommendation System which can utilize the power of machine learning and data analytics to help individual investors take better investment decisions. Conventional stock trading platforms are mostly tools for trading execution with minimal analytical support. Our project will focus more on predictive analytics and recommendations than trading. The methodology used for this project is a quantitative, data-driven one that makes use of past stock market data, technical indicators and sentiment analysis. It relies on the use of machine learning techniques and in particular Long Short-Term Memory (LSTM) models which enable you to predict the stock price direction. Data is obtained from trusted financial APIs like Yahoo Finance and Alpha Vantage, preprocessed, feature engineered, trained on... System is constructed on following systematic software development life cycle (SDLC) planning, designing, developing, testing and implementing. Agile methodology has been adopted so that iterative improvements are made based on the feedback and the performance conducted. Important results of this project are: the prediction accuracy, which shows how well the model can forecast the future; and the decision support, which shows how the model can help investors to decide if buy, sell or keep the stock. The results obtained proved that the combined use of technical indicators, sentiment analysis, historical data and machine learning models yield better results. This solution offers a user at a personal computer a solution through which to manage Buy, Hold or Sell actions to enhance their investment strategy decreasing risk. The implications of this project on the area of information systems are that of an application of artificial intelligence in a decision support system and financial analytics.

*Keywords: Artificial Intelligence, Stock Market Prediction, Machine Learning, LSTM, Financial Analytics, Stock Recommendation System, Data Analysis, Predictive Modeling, Sentiment Analysis, Investment Decision Support.*

## **Introduction**

### **Background and Context**

Financial markets are growing very fast and electronic trading systems are becoming more and more popular for traders, investors and institutions. Yet, even with trading tools at their disposal, investors still have difficulty in predicting the behavior of stocks, which is very complex, volatile and unpredictable. The traditional trading terminals just focus on the ability to buy and sell stocks, rather than providing tools and analytics for users to analyze trends and risks. Recent developments in Artificial Intelligence (AI) and machine learning present ample opportunities for improving decision-making in finance. Predictive modelling through the use of machine learning models would allow investors to interpret vast sequences of past stock data and project its future development. Compounding techniques such as technical indicators and sentiment analysis on financial news could then be applied to the data to further refine the information given to investors. Located in the discipline of information systems, this project deals with the use of the AI powered analytic system to design a smart stock recommendation system. Through employing the powerful newer generation data science methods, this project intends to connect the bank of raw financial data and the sound investment suggestions for their users.

### **Problem Statement**

Although there are thousands of stock trading applications available on the market today, there seems to be a lack of intelligent, data driven system to offer the accumulated knowledge to help users make better decisions. Today the existing stock trading tools mainly focus on executing transaction and make prediction by market trend. Investor will tend to rely on non-scientific and unreliable guess or external information while making decision. This will be dangerous to their investment. In addition, although there are some tools can make prediction, most of them either are not open or not easy to access or integration of many approaches such as technic indicators and sentiment index and etc. with

the requirement of having a simple, reliable and explicative recommendation web application. This project tackles the main issue of the absence of a unified platform that integrates AI forecasting and data analytics with user-friendly recommendations to provide an insight for stock investment decisions.

### **Objectives**

The main goal of this capstone project is to create an AI-Based Stock Market Analysis and Recommendation System that can provide complete investment recommendations and market analyses.

The specific objectives of the project include:

- Use the past two years to find a machine learning model, such as LSTM, with which to predict the tendency of a stock price.
- To apply technical analysis indicators (like moving averages, RSI etc.) in order to facilitate stock analysis.
- To combine sentiment analysis on financial news to better prediction accuracy.
- To design a recommendation system offering practitioner's valuable decision making knowledge in form of Buy, Hold or Sell actions.
- To create a user interface that is easy to use and displays analysis results clearly.
- To guarantee that the system focus mainly on analytics & recommendations(80%), with small percentage on trading facilities(20%).

### **Scope**

This project primarily aims to develop a data driven system that offers stock market analysis and suggestions based on AI and machine learning techniques. Scope of the project is limited to data collection, data processing, model development, integration of the model to the system and presentation of results.

The system will mainly make use of past data of stock markets from trustworthy financial APIs like Yahoo Finance and Alpha Vantage. It will also make use of forecasting, technical indicators and sentiment analysis.

Nonetheless this project has some limitations. This project is not set up to create one complete functioning real-time trading system that could perform live financial trading. Instead it would just aid in analysis. The accuracy of predictions may be affected by false readings generated by market volatility and limited data.

The project, overall, is devised to be a feasible approach within the academic budget and deadline, which is practically and scalable to show the application of AI in the above FIs.

## **Literature Review**

### **Relevant Theories and Models**

The theory and analytical model used in the development of an AI-based stock market analysis and recommendation system is based on the following theories in finance, machine learning, information system..

The first exemplar theory is the Efficient Market Hypothesis (EMH), claiming that stock prices are the best estimators of any information, therefore no consistent higher return would be realized by using any information. But evidences point out that there are over or under-value in the majority of the markets, especially in short-term dynamics, investors may still make profit after applying sophisticated statistical methods, which can be treated as Machine Learning tools (Fama, 1970).

Another concept is technical analysis which uses historical data of prices, charts, screens as well as technical indicators, like moving averages, RSI, Bollinger band, etc., in an attempts to predict future changes in the prices, which can be used as an input for automated trading systems (Murphy, 1999).

In an algorithmic context, machine learning models such as deep learning methods like Long Short-Term Memory (LSTM) models are increasingly useful in stock price prediction. LSTMs are highly

effective for a time-series task as they are able to learn long-range dependencies and patterns in sequential data. Studies suggest that LSTM models outperform classical statistical models in predicting stock prices as they are better able to discover complex nonlinear relationships (Fischer & Krauss, 2018).

Furthermore, Sentiment analysis has been gaining application in financial analysis. Through analysis of textual information obtained from news articles, social media websites and financial reports, sentiment analysis can contribute to financial analysis in evaluating the sentiment of the market and its potential relationship to stock prices. The inclusion of sentiment analysis and traditional quantitative models would improve predictive performance.

These theories and models together provide the theoretical foundation to the proposed system to have data driven analytical 'smarts' and intelligent recommendations.

### **Previous Work**

There has been considerable research conducted on the utilization of machine learning techniques and data analytics for stock market prediction and recommendation systems. The initial techniques used statistical methods such as linear regression and autoregressive integrated moving average (ARIMA) with limited ability to perform accurate forecasting and models complex nonlinear relationships.

In response to the developments in artificial intelligence, the use of machine learning methods (such as support vector machines (SVM), random forests, neural networks, etc.) has been more widely used for better prediction. Deep learning models (e.g. LSTM) have been specifically used.

Fischer and Krauss (2018) showed that LSTM networks outperform traditional approaches significantly when predicting stock returns, such is the power of the method in financial prediction. And for example, it was shown that the integration of domain knowledge in the form of technical indicators with machine learning method would have increased performance within a specific task.

Recently there have also been studies on the addition of sentiment analysis to stock prediction. Researchers have analyzed the sentiment of financial news and social media and observed that market sentiment has effect on stock movement that is mainly noticed in a short term. This has led to the development of hybrid models that use this information along with quantitative data.

However, many of the current systems still have some drawbacks, such as poor time or resource efficiency, unsatisfactory interpretability, poor integration of different data sources, focus on rather than recommendation output, or prediction with much less interpretability or learning advice output, and so on.

This project intends to fill these gaps by designing an integrated recommendation system, a unique platform integrating machine learning predictions, technical analysis and sentiment analysis. With focus on both usability and accuracy, the solutions will advance decision support systems in finance.

## **Methodology**

### **Research Approach**

This capstone project follows a quantitative research approach; since the emphasis is on analyzing stock market numerical data to recognize trends, as well as prediction power and forecast patterns. A quantitative approach research methodology is appropriate since the statistically and computationally-driven techniques will be possible for able to assess stock performance and provide data-informed results and suggestions.

The study approach is contrasted as a data driven analytical approach in which historical stock data are manipulated and analyzed using machine learning algorithms, so there is more objective, accurate and reproducible results while adopting quantitative indices in the evaluation of prediction accuracy, and mean squared error (MSE).



The methodology chosen fits the goal of the project to create an AI-based system to give trustworthy and explicable stock recommendations, by allowing the use of techniques of predictive modeling and financial analytics.

### **Data Collection**

Data collection is an essential process in this project because the quality of input data will affect the performance of the system. For this project, a secondary form of data is used and the data is available from several sources, including stock market historical data using Yahoo finance API and Alpha Vantage API.

The data gathered consist of general data, interesting data and text data. The general data comprised of stock prices( open, high, low, close), trading volumes, and the time series data on the stocks. Text data used, to derive sentiment analysis came from the financial press, and consisted of news items.

**Structured Collection of Data** The collection of the data was performed in an ordered manner so as to maintain the uniformity and completeness of the database. Use of authentic and popular data sources also aids in formulating reliability of the database.

### **Data Analysis**

The data analysis process is a very broad term, the several stages included are as follows: preprocessing, feature engineering, model development and evaluation.

Preprocessing of the collected data, such as cleaning, missing value handling, normalization and transformation into a format suitable for analysis is performed. The newly created dataset will then be subjected to feature engineering to derive and extract information (e.g. technical variables such as moving averages, RSI).

Our main analytical tool is an LSTM model to forecast the direction of movement on stocks out of the historical trend. LSTM is selected as the language model because of its special characteristic of modeling the temporal dependency.

Furthermore, sentiment analysis is also conducted on financial news data to measure different sentiments of the market. The outcome of sentiment analysis is combined with the predictive model and technical indicators in order to produce a robust stock scoring system.

Accuracy and mean squared error (MSE) are used as metric to check the performance of the model.

### **Justification of Methodology**

Quantitative research is most appropriate because of the problem context: the data required for research analysis is the numerical financial data, and the data for prediction models. Through use of Quantitative research, a more systematic, objective approach lends itself to data collection, implementation, and results evaluation of the solution.

The application of machine learning techniques, namely LSTM, is justified by similar work in the literature, showing how well they work for time-series forecasting or financial prediction. Also, the use of technical indicators and sentiment analysis is an advantage, as it is also based on other work.

The methodology we selected is relevant to our project by how it facilitates building an efficient, data driven and explainable recommender system. The methodology has its strengths such as scalability and precision. But it has its weaknesses as well such as its reliance on the quality of data and market volatility.

This approach is broadly effective in fulfilling project objectives and represents a valuable contribution to the realm of AI in financial analysis.

### **Project Plan and Execution**

#### **Project Plan (WBS and Gantt Chart)**

The project plan of the AI-Based Stock Market Analysis and Recommendation System was created with the use of structured project management tools such as Work Breakdown Structure (WBS) and Gantt chart.

The project has been broken down into a Work Break Down Structure using the goal-based method into five distinct phases: initiation, planning, execution, monitoring and closure. Specific tasks within each of these phases have been identified such as collection of data, pre-processing of data, programming of models, integration of analytics, presentation of results and design of user interface.

The Gantt chart aided in setting the project schedule and represented the order and time-frames for the tasks. It was used to model the dependencies between activities and plan a logical flow of work. E.g. the data had to be prepared prior to model creation and testing was performed after the system was in place. The Gantt chart also supported ongoing tracking and early problem detection.

Designed with the the WBS and Gantt chart, it was used as a working tool for planning, co-ordinate and control.

### **Execution (SDLC Implementation)**

The project is implemented based on the Software Development Life Cycle model (or SDLC). The SDLC stages used in our project are:

- Initiation Phase: Established the project scope, goals, and viability, and determined the requirements of the project.
- Planning Phase: Requirement analysis; Selection of tools and technologies; Development of project schedule and risk management plan.
- Design Phase: concentrated on the designing of the system architecture, such as data flow diagrams, model designing and analytical components integration.
- Development Phase: Included data gathering, preprocessing, designing the LSTM model, integrating the technical indicators and sentiment analysis.,

- Testing Phase: Involving measurement of system accuracy, mean squared error (MSE), usability, function.
- Implementation Phase: Developed the completed system prototype with user interface and recommendation engine.

The implementation of SDLC provided an organized and structured development process and made sure the theoretical concepts was turned into practical implementation.

### **Challenges and Mitigation**

Throughout the course of the project there were issues which arose which needed to be managed in order to get around the situation.

Data quality and inconsistency posed a significant challenge as stock market data received from APIs had a fair share of null/malformed values. This problem was addressed using various methods of data preprocessing that include data cleansing, normalization and validation.

One more challenge was the difficulty in the generation of machine learning models which were needed in the implementation of the learning algorithm (LSTM network). These models needed a lot of computing power and they needed to be fine-tuned. Therefore, a phased implementation was undertaken, that started with smaller sets of data and improved the parameters step by step.

-Time constraints were also a factor because of the short amount of academic time we have. This was managed by scheduling well using the Gantt chart and prioritizing the essential features until time ran out.

Lastly, some problems of integration complexity have been taken into consideration by means of modularization, implying the development and testing of various parts separately and connection of them afterwards:

In general, such mitigation strategies were successful in keeping the project moving while maintaining the high quality.

## **Findings and Analysis**

### **Data Presentation**

The data for this project was comprised of historical stock market data, obtained from authoritative financial data providers, namely Yahoo finance and Alpha vantage. This data set had various fields of interest, which includes; open, high, low, close and volume.

The data was initially preprocessing for accuracy and consistency. This was to treat missing data, normalize the dataset and convert the data for machine learning, formus. Key visualizations, such as line graphs and market trend charts were used to graph stock prices against time to identify any patterns and trends in stock values.

Besides news, several technical indicators were also computed and plotted, for example moving averages and relative strength index (RSI). This type of visual visualization can help further interpret the trend, identify overbought or oversold markets, or suggesting possible reversal points.

### **Analysis**

The main body of analysis was carried out using a single Long Short-Term Memory (LSTM) model, trained on historical time-series data, and tested based on accuracy and Mean Squared Error.

From the analysis, it is clear that the LSTM model is powerful enough to identify trends and predict stock prices quite accurately. The results seem to indicate that technical indicators enhance the potential for the LSTM mode to make predictions.

In addition to the system, sentiment analysis was used by analyzing financial news. The results demonstrate that positive sentiments usually lead to rising price and negative sentiments are inclined to falling. This emphasizes that the significance of qualitative data should be revealed.

This hybrid approach, using the predictive modeling, technical analysis and sentiment analysis, increases the overall accuracy of the system.

### **Implications**

A final conclusion that can be drawn from the results of this project is that the use of AI and Financial Analytics helps to considerably simplify stock market decisions. The system developed offers users mental shortcuts to decision making in the form of Buy-Hold-Sell signals.

Another implication of this project from the information systems point of view is understanding how all these different data and techniques were combined in order to produce a decision support system that would provide sound recommendations, which was achieved here by the use of machine learning classifiers such as LSTM.

However, the results show that stock market prediction can be unreliable because of market instability and various influences. So, the prediction has only reference value rather than a conclusive outcome.

Generally, within the project, the work done is a development of the fields concerned and the overall contribution to the field of study.

## **Discussion**

### **Evaluation of Outcomes**

This project has shown how combining of artificial intelligence and financial analytics can assist decision making on stock market. The published system provides a working example of how the model is constructed using LSTM, as well as the use of technical indicators and sentiment analysis to produce stock recommender.

From the above results it shows that the system is able to recognize trends and predict quite accurately on its own using history data, further adding technical indicators increases the effectiveness of the system and sentiment analysis increases the understanding of the state of the market for the system.

Where the evaluation of results brings out possible weaknesses and shortcomings: the accuracy of the prediction is depending on data quality and availability, and the model could not be viewed as

complete with regard to unforeseen events, as economic or political changes occur. Nevertheless, the tool does meet its initial goal of delivering an effective decision support.

### **Comparison with Existing Literature**

The results from the study are in agreement with the existing literature relating to the application of machine learning models, especially LSTMs, on time-series forecasting in financial markets. This previous research establishes that LSTM models outperform traditional statistical methods on prediction accuracy.

In the same way, using technical indicators is also in accordance with the traditional ways of analyzing market trend and movements. Furthermore, by using sentiment analysis the system builds on the recent research trends, which explore the effects of external sources of information in stockprice movements.

However, existing systems are often limited to prediction with no concrete recommendations. Our project builds on existing work by integrating a predictive system with a recommendation engine that is relatively simple and easy to use while allowing for explanations to be made about the recommendations - an important gap identified in the literature.

### **Limitations**

Although the project is fairly good, there are some limitations. Most obviously, historical data cannot absolutely guarantee what will happen in the markets in the future.

An additional drawback is the high computational complexity of the machine learning models, like LSTM, who needs lot of time and computational power for training. This can limit its use in real time applications.

Also, sentiment analysis can be limited by the nature of the text and the accuracy of sentiment classification may impact the overall prediction results. Moreover, as the project can be developed for both analysis and trading, this project does not implement live trading.

These limitations serve as indicators of failure and guide us as to where improvements can be made and where future researchers can focus their efforts.

## **Conclusion**

### **Summary of Findings and Deliverables**

The final outcome of this capstone project was a well implemented AI-Based Stock Market Analysis and Recommendation System that combines machine learning, technical analysis and sentiment analysis to assist investment decision making. This also filling a significant gap of stock market platforms to not only support transactions but to explain and back the investment decision.

The results show that applying LSTM models to time-series stock data is feasible and can be used to predict the trend of prices. Results show that the model did a fairly good job of detecting trends in past data and producing reasonable predictions. This system can have an additional layer of analytical insight if used with technical indicators like moving averages, RSI etc.

In addition, the use of sentiment analysis improves the prediction power of system by encompasses the effects of the external elements such as financial news and market sentiment. It is implied that it is better to merge quantitative prediction with qualitative determination in financial analysis.

The project stresses on usability as well by converting the complex analytical results into simplified clear recommendations like Buy/Hold/Sell that can be easily understood by the user and could be effectively used.

The key deliverables of this project include:

- A ready-to-use functional prototype of this AI based stock analysis and recommendation system
- A machine learning model (Long short-term memory, LSTM) for stock prediction
- Implementation of technical indicators for market analysis
- Incorporation of sentiment analysis for further accuracy in prediction



- A recommendation engine (Buy/Hold/Sell recommendations)
- A graphical user interface to display stocks trends and analytics
- All documentation relating to the project, including project methodology, results, discussions etc.

In conclusion, the project has met the objectives by showcasing how AI and data analytics can be utilized in the financial decision making process. It provides a base that can be improved upon for future iterations and makes a meaningful contribution to the information systems literature.

### **Recommendations**

From the evidence and analysis concluded regarding the AI Based Stock Market Analysis & Recommendation System; there are various recommendations generated as a result of the research undertaken.

Secondly, it is desirable to use other data types, such as market data and macroeconomic variables (interest rate, inflation, world markets trend..). Using more types of data will help reveal the real market behavior and forecast more accurately.

Second, the performance of the machine learning model could be optimized with a wide variety of optimization techniques, such as hyperparameter tuning, cross validation, and use of ensemble learning techniques. For instance, in building ensemble learner combining a growing number of learners could improve the predictive outcome and yet not increase the risk of overfitting.

3) Improve sentiment analysis: develop new algorithms to analyze the sentiment of the news and social media users. These algorithms should utilize more advanced natural language processing techniques and deep learning-based models.

Furthermore, the user interface should be expanded and the visualization features should be improved. Interactive dashboards with intuitive graphical representation and easier recommendations outputs will make the system more user-friendly.

A further important suggestion is the addition of risk analysis tool such as measures of volatility and risk scoring in order to help users to make investment decisions more informed. This complements the recommendation system by showing the risks of the stocks.

Finally, incorporating live market monitoring and automated alerts in future system iterations could prove beneficial, so users could be prompted in advance of market changes and recommendation variations.

### **Future Work**

Though the AI Based Stock Market Analysis and Recommendation System clearly illustrates that how machine learning and analytics can be successfully applied for investment decision making support, some further improvements and extensions could be:

One of the significant directions that the system can be developed in the future is real-time data processing. At present, the outputs are based on existing data. In real applications, if they can obtain data streams directly from the market, the system would be more useful to give more accurate and timely prediction and suggestions.

Another avenue for improvement could be the implementation of advanced machine learning architectures, or hybrid solutions. Efficient algorithms such as reinforcement learning, transformers-based architectures, or ensemble models could be implemented in future works to boost accuracy and responsiveness to market fluctuations.

Another improvement to the system can be made by supplementing the sentiment analysis with a better classifier. Technologies like Text Classification using Deep learning[9], or language models in natural language processing will for sure improve the sentiment analysis.

Furthermore, a future portfolio could be created that is fully integrated into a trading system, enabling users to trade open orders on system prediction alone. Although attractive, certain issues will need to be addressed including security, compliance and risk management.

Regarding another scope of development, portfolio management functionalities are to be added so that the users will be able to efficient manage multiple stocks, view portfolio performance and choose the best combination of stocks depending upon the perceived risk return trade-offs.

Lastly, optimizing the user interface and visualization will improve usability and accessibility. Creating mobile apps or live dashboards can enable users to have better interaction with the system.

To conclude, these are the major proposed modifications that will optimize the system and make it even more scalable, accurate, and applicable to real life.

## **Appendices**

### **Appendix A: Project Plan (WBS and Gantt Chart)**

This appendix includes the project planning artifacts I used in order to successfully develop the AIBased Stock Market Analysis and Recommendation System. The following are the Work Breakdown Structure (WBS) and the Gantt Chart.

The Work Breakdown Structure (WBS) is done into a form of the project divides by project key phases like initiation phase, planning phase, implementation phase, monitoring and controlling, and closing phase, and each phase further divides into several task like collecting data, pre-process, build model, analytics implementation, and system testing in order to clearly define the project scope and responsibilities.

The Gantt Chart presents the full nature of the project schedule from start to completion. The time scales and task dependencies are clearly visible and clearly define the order in which tasks such as data pre-processing and modeling take place. During execution, the Gantt Chart may be updated to monitor progress.

The WBS and Gantt Chart work in conjunction as a vital tool of project planning and management to keep the project organized, efficient and be in accordance with project objectives.

### **Appendix C: Additional Data and Analysis**

This appendix contains additional data and analysis outputs for the results discussed in the main report. It contains detailed information on the data processing, model results and what data analysis techniques were used to build the system.

The appendix presents larger datasets like historical stock prices, which were used as dataset to train and test the machine learning model. More graphs like stock graph with moving averages, stock graphs with technical indicators graphs were added that could not be presented in the main report.

In addition to the performance metric used to evaluate the models, comprehensive results of the implemented LSTM model are provided. This may include, but are not limited to MSE, prediction comparisons and detailed application results.

Also present is further analysis arising from the sentiment analysis, illustrating negative and positive sentiment scores in relation to stock movement. This shows evidence of qualitative data being incorporated into the analysis.

Overall, this appendix supports the analysis and findings of this report with detailed evidence, and helps to improve the transparency and integrity of this work.

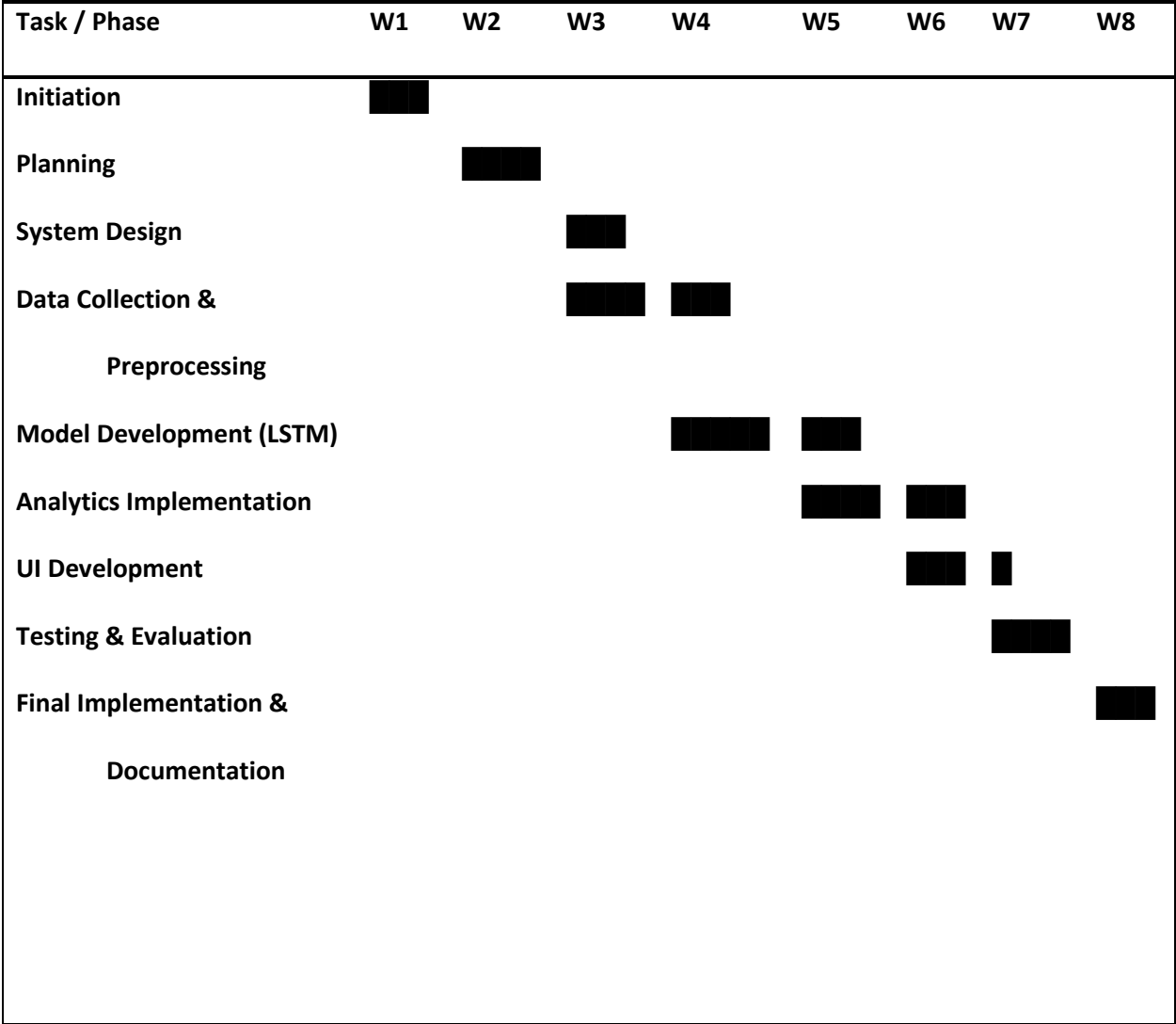
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Tables

Table 1

Gantt Chart for AI-Based Stock Market Analysis System



*Note:* A Gantt chart shows a graphical presentation, how the activities of AI-Based Stock Market Analysis and Recommendation System project will carry out in sequence and schedule. The Gantt chart describes the schedule of activities and the relationship among them, all identified activities like start up, Planning, system design, Data collection, model building, analytics,

implementation and so on are scheduled; helps to control the progress of project, identify dependencies of activities and meet the target of project within schedule of time.